复旦大学化学系

2006～2007 学年第一学期期末考试试卷

□ A卷

课程名称：__Physical Chemistry__ 课程代码：_____

开课院系：__ Chemistry _ 考试形式：闭__卷__ 成绩：____

姓 名：_____ 学 号：_____ 专 业：____

No.	1	2	3	4	5	6	7	8
Score								
No.	9	10	11	12	13	14	15	16
Score								

This is a closed book exam. Use of a calculator and an English dictionary is permitted. Show all of your work and check your units carefully. **Don't give help to, or get help from, others.** Thanks for your cooperation. **GOOD LUCK** !

Some useful constants and results are:

(atomic unit)

Relative atomic mass: Ca-40 O-16

1. Work functions, , are usually listed in units of electron volts, , where 1 = . What is the velocity of an electron emitted by Li ( ) if light having a frequency of is absorbed? （4 points）

2. Calculate the de Broglie wavelength of an electron traveling at 1% of the speed of light ( ). (4 points)

3. Which of the following expressions are acceptable wavefunctions, and which are not? For those that are not, state why? （8 points）

1) , ; 2) , ;

3) , ; 4)

4. Using Bohr interpretation, for an electron having a one-dimensional wavefunction in the range to 1,

1) Normalize the wavefunction;

2) Calculate the probability that the electron is in the first half of the range, from to ;

3) Calculate the probability that the electron is in the middle half of the range, from to .

4) Calculate the energy of this electron (assume that the potential energy is zero, or).

5) Calculate the average value of the position of this electron in the range of to 1. (12 points)

5. -Carotenes （ -胡萝卜素） are highly conjugated polyenes found in many vegetables. They can be oxidized and used to synthesize pigments that play important roles in the chemistry of mammalian vision. The parent compound, -carotene, has a maximum absorption of light that occurs at 480 . If this transition corresponds to an to transition of an electron in a particle-in-one-dimension-box system, what is the approximate length of the molecular “ box ”(6 points)

6. What is the average value of the total angular momentum for for the hydrogen atom? （5 points）

7. For a body-centered cubic lattice, please calculate the inter-planar spacing between the (100), (110) and (111) planes, respectively. (6 points)

8. For two the hybrid orbits on the same atom , and ,

1) Show that they are orthogonal;

2) Show that they have same energy.

3) Show that they are oriented 120 ? apart. (12 points)

9. Predict the bond order in each of the following species:

1) N_2^+; 2) H_2^-; 3） Li_2^+; 4) C_2^+

10. A metallic solid is characterized by a unit cell with an atom of element X located at each of the corners of the cell, one atom of element Y at the center, and an atom of element Z located at each of the edges. Please give the empirical formula of the solid. (5 points)

11. The ionic radius of Ca^{++} and O^{2-} are 100pm and 140pm, respectively. (12 points)

1) Which kind of coordination polyhedra does calcium oxide belong to?

2) If adjacent Ca^{2+} and O^{2-} spherical ions are touching , calculate the edge length of the unit cell;

3) Calculate the density for this solid.

12. The valence electron configuration of gold is $5d^{10}6s^1$. Recent studies show that the bonding property of gold is quite similar to hydrogen. Please using Huckel molecular-orbital theory to determine whether the gold trimercation (Au_3^+) is a stable molecule or not. If it is stable, please tell the most stable structure of the cation. (12 points)

13. Both CO and H_2O molecules can be used as ligands. It is found that iron atom can coordinate five CO molecules to form $Fe(CO)_5$, which is quite stable. However, no stable $Fe(H_2O)_5$ molecule has been observed. Why?(6 分)

14. What different point groups may the biphenyl molecule belong to, depending on the rotational relationship of the two rings about the central C-C bond? Consider the same question for ferrocene. (6 points)

15. For a bent AB_2 molecule, taking the p orbital perpendicular to the molecular plane of each B atom as bases, work out the symmetry-adapted linear combination orbitals using projection operators (8 points).

C_{2v}	E	C_2	σ_v(xz)	σ_v(yz)		
A	1	1	1	1	z ,	x^2, y^2 ,z^2
A_2	1	1	-1	-1	R_z	xy
B_1	1	-1	1	-1	x,R_y	xz
B_2	1	-1	-1	1	y ,R_x	yz